

Introduction

The Ghent PDF Output Suite was created for users who process PDF files in the graphic arts industry, as an aid to determine whether their workflows are behaving conforming the PDF/X standards. These patches can be used by end users of graphic arts equipment as well as developers of applications that handle PDF files.

The suite consists of a series of PDF patches. Each patch tests a specific property of a PDF/X file. The patches can be used on their own but the intention of the suite is that the patches are grouped together (as PDF pages would normally be grouped together within a workflow). It is likely that application settings and RIP settings can have a significant effect on the results.

All the issues tested by these patches are real world issues that can be found in a production environment; however, these patches do not reflect normal production files and the results may in some cases appear extreme. They have been carefully constructed to allow effects that are normally subtle to be seen clearly and unambiguously and this should be taken into account when evaluating the results of any tests based on these patches. On a technical note, all patches conform to either the PDF/X-1a, PDF/X-3 or PDF/X-4 ISO standard but they do not always conform to the Ghent Workgroup PDF/X-Plus specifications.

It is likely that this suite will be updated, new patches will be added, and existing patches will be revised. For this reason the documentation for each individual patch is distributed along with the patch. It is advised to check regularly for updates to the suite on the Ghent Workgroup website at <http://www.gwg.org>.

Version 5.0 Release Notes

In addition to the patches from version 4.0 in version 5.0 additional patches have been added to test ICCbased objects allowed in PDF/X-4 (ISO 15930-7) in order to test device independent workflows. We have abstained from using ICC based blending spaces in isolated transparency groups since their processing is not clearly defined in PDF 1.6 (base of PDF/X-4).

Contributors

The following members of the Ghent Workgroup have actively contributed patches to version 5.0 of the Ghent PDF Output Test Suite:

- Didier Haazen, VIGC, Flemish Innovation Center for Graphic Communication (B)
- Peter Kleinheider, inpetto (A), representing PDFX-ready
- Stephan Jaeggi, PrePress-Consulting (CH), representing VSD

Delivery

The patches from the Ghent PDF Output Suite V4.0 are also included in the package. The patches are divided into three groups:

- **CMYK**: test patches using DeviceCMYK color only.
- **SPOT**: test patches using DeviceCMYK and a spot color.
- **ICC-CMS**: test patches using ICCbased colors and testing color management.

CMYK

• GWG 1.0	CMYK Overprint Test	PDF/X-3	old
• GWG 1.1	CMYK Overprint Mode	PDF/X-3	old
• GWG 4.0.1	White Overprint Patch	PDF/X-1a	old
• GWG 5.0	Font Substitution	PDF/X-3	old
• GWG 5.1	Font subset and substitution	PDF/X-1a	old
• GWG 5.2	Font subset and substitution	PDF/X-1a	old
• GWG 6.0	Use of shadings	PDF/X-1a	old
• GWG 6.1	Use of shadings	PDF/X-1a	old
• GWG 9.0	Font Support	PDF/X-3	old
• GWG 9.1	OpenType Font Support	PDF/X-4	old
• GWG 15.0	Optional Content (OCCD) *	PDF/X-4	old
• GWG 15.1	Optional Content (RBGroup) *	PDF/X-4	old
• GWG 15.2	Optional Content (OCMD) *	PDF/X-4	old
• GWG 16.0	Transparency Blend Modes (DeviceCMYK, Non-Knockout)	PDF/X-4	old
• GWG 16.1	Transparency Blend Modes (DeviceCMYK, Knockout)	PDF/X-4	old
• GWG 16.2	Transparency Blend Modes (DeviceCMYK, Isolated)	PDF/X-4	old
• GWG 16.6	Image Softmasks (DeviceCMYK)	PDF/X-4	old
• GWG 16.8	Vector Softmasks (part 1)	PDF/X-4	old
• GWG 16.9	Vector Softmasks (part 2)	PDF/X-4	old
• GWG 16.10	Text Softmasks (part 1)	PDF/X-4	old
• GWG 16.11	Text Softmasks (part 2)	PDF/X-4	old
• GWG 17.0	JPEG2000 compression (DeviceCMYK)	PDF/X-4	old
• GWG 17.3	JBIG2 compression	PDF/X-4	old
• GWG 18.1	16Bit Image (DeviceCMYK)	PDF/X-4	new image
• GWG 19.0	DeviceN Overprint (Black)	PDF/X-1a	old
• GWG 19.1	DeviceN Overprint (Yellow)	PDF/X-1a	old
• GWG 19.2	DeviceN Overprint (White)	PDF/X-1a	old

* The patches for optional content cannot be combined with the other patches for optional content in one PDF file.

SPOT

• GWG 2.0	Spot to CMYK Overprint	PDF/X-1a	updated
• GWG 3.0	Gray Overprint Patch	PDF/X-1a	old
• GWG 3.1	Gray Image Overprint (CMYK over Spot)	PDF/X-1a	old
• GWG 4.1	White Overprint Mode	PDF/X-3	old
• GWG 8.01	DeviceN Support (6 colors)	PDF/X-3	old
• GWG 8.1	DeviceN Support (5 colors)	PDF/X-1a	old
• GWG 8.2	DeviceN Support (4 colors)	PDF/X-3	old
• GWG 12.0	White Overprint/Knockout	PDF/X-1a	old

ICC-CMS

• GWG 13.0	ICC Source Profile	PDF/X-4-ICC	new
• GWG 13.2	ICC based CMYK Overprint	PDF/X-4-ICC	new
• GWG 13.3	ICC based RGB Overprint	PDF/X-4-ICC	new
• GWG 16.1	Transparency Blend Modes (ICCbasedRGB)	PDF/X-4-ICC	new
• GWG 16.4	Transparency Blend Modes (ICCbasedCMYK)	PDF/X-4-ICC	new
• GWG 16.7	Image Softmasks (ICCbasedRGB)	PDF/X-4-ICC	new
• GWG 17.2	JPEG2000 compression (ICCbasedRGB)	PDF/X-4-ICC	new
• GWG 18.0	16bit Image (ICCbasedRGB)	PDF/X-4-ICC	new
• GWG 18.2	16bit Image (ICCbasedGray)	PDF/X-4-ICC	new
• GWG 18.3	16bit Image (DeviceGray)	PDF/X-4	new
• GWG 18.4	16bit Image (ICCbasedCMYK)	PDF/X-4-ICC	new
• GWG 20.5	CMYK Image with ICC V4 profile	PDF/X-4-ICC	new
• GWG 20.6	RGB Image with ICC V4 profile	PDF/X-4-ICC	new
• GWG 22.0	Color Conversion Indicator	PDF/X-4	new
• GWG 22.1	OutputIntent Change Indicator	PDF/X-4-ICC	new
• GWG 23.0	Four different Grays	PDF/X-1a	new

The label **PDF/X-4** indicates that this patch uses PDF/X-4 features. The label **PDF/X-4-ICC** indicates that this patch uses ICC based colors which must be color converted for the output.

Each of the patches is documented in an corresponding **ReadMe** document.

NOTE: The numbering of the patches is not continuous since some numbers are reserved for planned patches for the ICCbased workflow.

Test pages with assembled patches

The file **Ghent_PDF-Output-Test-V50_ALL_X4.pdf** contains six A4 size pages (which can be cropped to Letter size pages) with 48 GWG test patches for easy testing of PDF/X-4 workflows. The test pages are also divided into the three categories: CMYK, SPOT and CMS. The folder of each category also contains the corresponding test pages.

These test pages should be processed like regular PDF/X-4 print jobs (e.g. imposed to a sheet for offset printing, printed as single or double pages in digital printing or placed like ads on a page). The pages contain test patches with essential PDF/X features. Errors are identified either with a cross or by a deviation from the integrated reference images.

The test patches have been assembled with Acrobat Pro and imposed with Callas pdfToolbox 9. This tool places each patch as isolated transparency group on the page. Other tools (e.g. imposition applications or ad layout applications) may do it differently.

The PDF has been verified for PDF/X-4 compliancy in Callas pdfToolbox 9, Enfocus PitStop 13 and Acrobat DC. The result of this preflight check is embedded as Preflight Audit Trail (a GWG standard!) into the PDF. The Preflight Audit Trail can be displayed in Adobe Acrobat/Reader 9 and higher with the menu [View > Navigations Panels > Standards](#). After embedding the Preflight Audit Trail the PDF has automatically been signed with a preflight signature. (In order to validate the preflight signature, the signature certificate must be added to the Trusted Identities via the signature properties in the signature panel of Acrobat). In case of an (unwanted) change of the PDF this signature becomes invalid. You can then go back to the original version in the signature panel. The preflight signature has no restrictions and does not prevent processing the PDF in a prepress workflow system!

NOTE: Although all patches are PDF/X-4 compliant, most of the patches are not compliant with the Ghent PDF Specifications. The reason is that for most patches special properties of the objects (e.g. overprinting) had to be used in order to clearly visualize a failure of the patch.

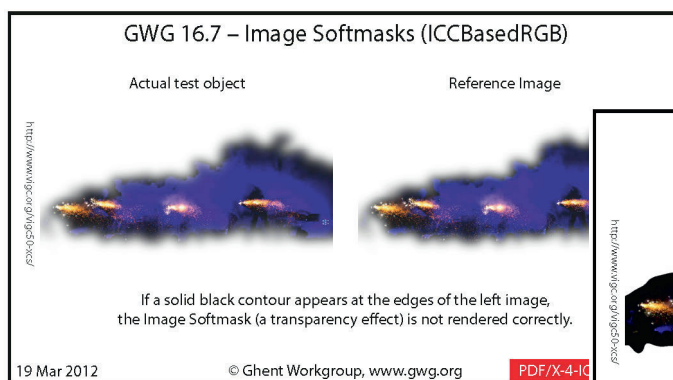
Evaluation of the results

We have tried hard to make the evaluation of the test results as easy and unambiguous as possible. (Hint: you will not need a loupe to evaluate the results.)

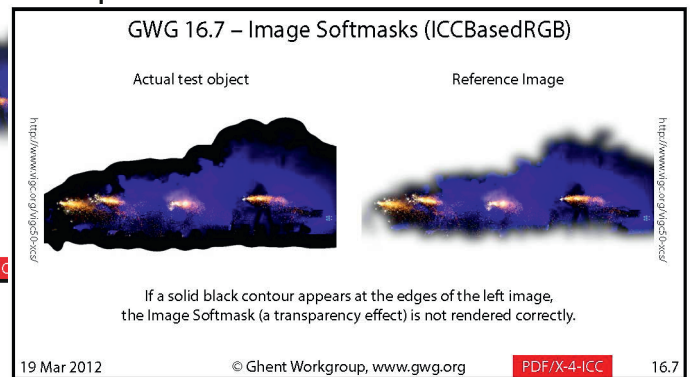
Two different methods are used to indicate errors:

Reference images

Where it is not possible to indicate errors with a cross, reference images are used to show the correct result:



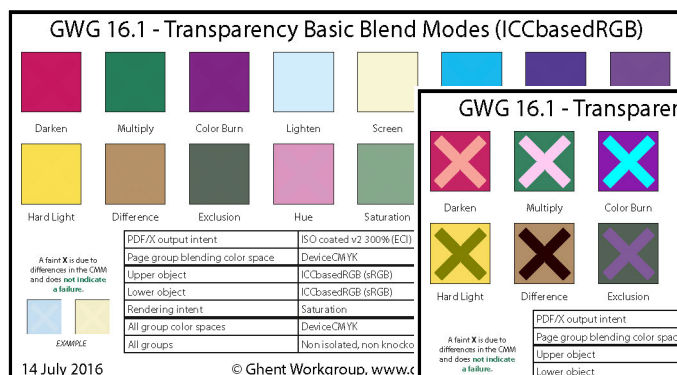
Correct rendering of the patch: the test passed. The Soft Mask looks the same as the reference image.



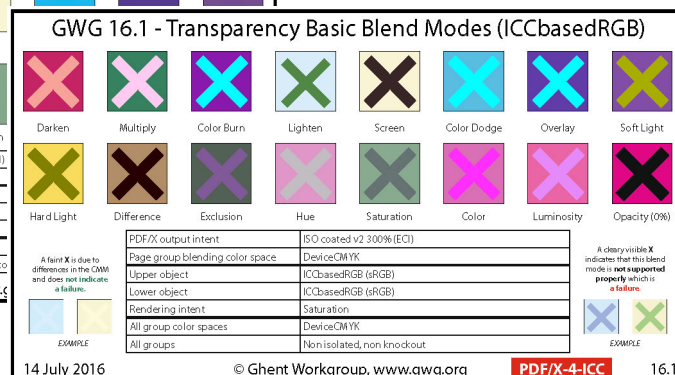
Incorrect rendering of the patch: the test failed. A clear black stroked fill is visible at the edge of the image instead of a smoothly softened edge.

Graphic indicators like cross or check mark

If possible errors are indicated with a clearly visible cross:



Correct rendering of the patch, all tests passed. No clear X is showing.



Incorrect rendering of the patch, all tests have failed. Each X indicates an incorrectly rendered test.

Notice

The most important design goal for these patches is the ability to identify an error clearly. The rule of thumb is to examine the patches with a viewing distance of 0.5 m / 20 inches. If you see a **clear X** then there is a **failure**. For some patches a **faint X** may appear due to rounding of values or different color management engines being used. This is **not a failure** of the tested features:



A **clear X** indicates a **failure** of this feature.

A **faint X** is **NOT a failure** of this feature.

If errors occur then the settings of the PDF output workflow might be wrong. Unfortunately still not all PDF workflow systems ship with PDF/X compliant settings. Many workflows are still changing the overprinting definitions of a PDF file. This is not allowed in a PDF/X workflow because the result the creator (customer) has seen (and accepted) in Adobe Acrobat/Reader or on a proof will be different from the final printed product if such changes are applied in a late stage. Therefore it's also highly recommended to output the Ghent PDF Output Suite test pages on a proofing device to make sure that the proof will represent the final printed matter.

Documentation

All new patches feature a caption with a short description of the objects and an explanation how a failure can be identified. In addition a ReadMe document with more detailed information is available for each (group of) patch(es).

Free download

The Ghent PDF Output Suite can be downloaded for free at <http://www.gwg.org/workflow-tools/test-suites/ghent-output-suite/>. Please check regularly for updates.

Feedback and questions

Please send your feedback and questions to info@gwg.org.

The fine print

Disclaimer

The patches of the Ghent PDF Output Suite contain only tests for basic PDF/X-4 features. By combining such features in complicated layouts new situations (e.g. nested transparency groups) can occur which are out of scope of these simple test patches.

Legal Notice

Use of the Ghent PDF Output Suite (which is defined as the totality of its patches, test pages and documentation files) is subject to the following conditions which are deemed accepted by any person or entity making use hereof.

Copyright Notice

Copyright © 2016, Ghent Workgroup (<http://www.gwg.org>). All Rights Reserved. The Ghent Workgroup hereby grants permission to use this test suite and its documentation as described in the associated documentation, subject to the following conditions. This legal notice must be included in all copies containing the whole or substantial portions of the Ghent PDF Output Suite. Without express written permission of the Ghent Workgroup it is not permitted to use this PDF Output Suite for anything but its intended purpose of testing workflow setup. The Ghent PDF Output Suite cannot be sold or used in any commercial context without previous written permission by the Ghent Workgroup.

The Ghent Workgroup and Ghent PDF Output Test Suite names are copyrighted by the Ghent Workgroup. All other names are product names, trademarks or registered trademarks of their respective owners and are hereby acknowledged as such.

Waiver of Liability

The Ghent PDF Output Suite is provided as is, without warranty of any kind, express, implied, or otherwise, including but not limited to the warranties of merchantability, fitness for a particular purpose and noninfringement. In no event will the Ghent Workgroup, the authors of the patches, or their employers be liable for any claim, damages or other liability, whether in an action of contract, tort or otherwise, arising from, out of, or in connection with the Ghent PDF Output Suite.